Learning Internal Alignment Embeddings for Scalable Multi-Agent Coordination

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Abstract

We present ESAI-v3, a reinforcement learning architecture that trains agents with differentiable *internal alignment embeddings* (IAE)—learned latent regulators that predict and attenuate externalized harm during multi-agent coordination. Unlike auxiliary reward shaping, IAE dynamics are supervised counterfactually, propagate via similarity-weighted graph diffusion, and provide interpretable bias-mitigation controls. In evaluations across five diverse environments (including Multi-Agent Particle Environments, Overcooked-style coordination, and Sequential Social Dilemmas), ESAI-v3 exhibits higher cooperative action rates, reduced alignment penalties, and demonstrates zero-shot scaling from 4 to 16 agents with 89% performance retention—compared to 52% for standard RL baselines. Ablation studies confirm causal necessity of all architectural components, and interventional probes establish that IAE magnitude causally regulates behavior beyond passive correlation (29% prosocial drop under suppression, $p < 10^{-4}$). Targeted bias-suppression experiments show controllable fairness-performance tradeoffs (42% bias reduction with <4% cooperation loss). We ground IAE semantics empirically via Pearson correlation $r = 0.74$ ($p < 0.001$) with environment-specific harm metrics. These results indicate that learned internal alignment embeddings can provide intrinsic coordination pressure in multi-agent systems; we discuss limitations, failure modes, evaluation scope, and governance requirements for real-world deployment.

1 Introduction

Contemporary multi-agent reinforcement learning (MARL) optimizes explicit task objectives but typically lacks internal differentiable regulators that encourage prosocial behavior and stable coordination under distribution shift. Standard approaches to alignment—such as reward shaping, constrained optimization, or inverse RL—require external supervision, careful hand-tuning, or access to human preference data (Hadfield-Menell et al., 2016; Christiano et al., 2017).

We investigate whether agents can instead learn an *internal alignment embedding* (IAE): a differentiable latent variable that tracks predicted externalized harm and shapes policy gradients toward harm reduction.

Motivating challenge: zero-shot population scaling. Consider training cooperative agents in small groups (4 agents) and deploying them in larger populations (16 agents) without retraining. Standard MARL policies degrade sharply—often by 40–50%—due to cascading negative externalities and coordination collapse (Foerster et al., 2018). We demonstrate that ESAI-v3 retains 89% of its coordination performance under this structural distribution shift, compared to 52% for vanilla policy-gradient methods and 63% for reward-shaping baselines. This zero-shot scaling result suggests that IAE-driven policies learn more robust coordination principles rather than overfitting to specific population structures.

Contributions. We introduce ESAI-v3, integrating four differentiable mechanisms:

1. **Differentiable counterfactual alignment penalty** via softmin reference distributions, enabling gradient-based harm forecasting (Sec. 4.3).
2. **IAE-weighted attention** biasing perceptual salience toward alignment-relevant features (Sec. 4.5).
3. **Hebbian affect-memory coupling** with differentiable read operations supporting temporal credit assignment (Sec. 4.6).
4. **Locally damped graph diffusion** with similarity-weighted propagation and bias-mitigation controls (Sec. 4.7).

We validate on five environments spanning moral temptation, social diffusion, cooperative navigation (MPE), human-compatible coordination (Overcooked), and tragedy-of-the-commons dynamics (SSD). Key empirical findings include:

- **Grounded semantics:** IAE magnitude correlates with ground-truth harm metrics ($r = 0.74 \pm 0.06$, $p < 10^{-3}$), addressing circularity concerns (Sec. 7.2).
- **Causal regulation:** Interventional probes show that manually suppressing IAE reduces prosocial ratio by 29% ($p < 10^{-4}$), confirming causal impact beyond passive correlation (Sec. 7.3).
- **Architectural necessity:** Pairwise ablations show no two-module subset recovers full performance (Sec. 7.4).
- **Controllable fairness:** Similarity-suppression regularizer reduces in-group bias by 42% with <4% cooperation loss (Sec. 7.6).
- **Interpretable attention:** IAE-weighted attention concentrates 2.4× more weight on victim-related features during high-harm states (Sec. 7.7).

Scope and limitations. IAE is a computational abstraction—*not* a claim about subjective emotion or consciousness. All experiments use simulated environments with relatively low state dimensionality; extension to high-dimensional robotic control remains future work. Human evaluation and domain-specific safety audits are required before real-world deployment. We discuss failure modes, bias risks, evaluation scope, and governance requirements in Sec. 8.

2 Related Work

Affective and cognitive architectures. Classical symbolic appraisal models (Ortony et al., 1988; Picard, 1997) lack differentiability. Modern continuous affect models (Broekens et al., 2013; Hofer and Smith, 2022) modulate reward magnitude but do not enable bidirectional gradient coupling between latent states and policy parameters. ESAI-v3 differs by treating the IAE as a first-class learned variable with explicit counterfactual supervision.

Intrinsic motivation and alignment. Intrinsic rewards drive exploration (Pathak et al., 2017) but rarely encode normative constraints. Alignment methods learn from human feedback (Christiano et al., 2017) or infer cooperative objectives (Hadfield-Menell et al., 2016). Multi-objective RL (Hayes et al., 2022) optimizes Pareto frontiers but lacks interpretable internal dynamics. ESAI-v3 combines counterfactual forecasting with differentiable graph diffusion to provide both alignment pressure and auditability.

Distinction from potential-based shaping. Potential-based reward shaping (Ng et al., 1999) adds $\Phi(s_{t+1}) - \Phi(s_t)$ to rewards, preserving optimal policies under certain conditions. ESAI-v3 differs in three critical ways:

1. **Latent dynamics:** IAE evolves via learned dynamics and graph diffusion, not hand-designed potentials.
2. **Counterfactual supervision:** $\widehat{E}_{t+1}^{(a)}$ is trained to predict harm outcomes, not task rewards.
3. **Attention coupling:** IAE gates perception (Eq. 13), enabling non-Markovian salience modulation unavailable to potential-based methods.

Potential shaping is a degenerate case where $E_t = \Phi(s_t)$ with no learning, multi-agent propagation, or attention mechanisms.

Multi-agent coordination and communication. Message-passing architectures (Foerster et al., 2018; Lowe et al., 2017) enable information sharing but lack normative grounding. Graph neural MARL (Agarwal et al., 2020) models relational structure; we extend this with similarity-weighted diffusion and bias-suppression controls. Counterfactual MARL methods (Foerster et al., 2018) attribute credit but do not maintain persistent internal alignment states with interpretable attention mechanisms.

Social value alignment. Recent work on social value networks in goal-conditioned RL (Yang et al., 2020) learns to predict and align with partners' objectives. ESAI-v3 differs by: (1) using counterfactual harm forecasting rather than explicit value prediction, (2) propagating alignment signals via graph diffusion with auditable similarity weighting, and (3) providing interpretable attention mechanisms that reveal which features drive alignment-relevant decisions. No existing work combines all four components (counterfactual regret, attention gating, Hebbian memory, graph diffusion) in a single differentiable architecture with demonstrated causal impact via interventional probes.

Differentiable memory. Differentiable memory systems (Graves et al., 2016) and Hebbian plasticity (Miconi et al., 2018) improve credit assignment. ESAI-v3 couples Hebbian traces to IAE dynamics, enabling alignment-aware memory updates while preserving end-to-end differentiability for counterfactual forecasting.

3 Architecture Overview

ESAI-v3 augments a standard policy-gradient agent with an internal alignment embedding $E_t \in \mathbb{R}^k$. All computations involving E_t are differentiable, enabling gradient propagation into policy parameters θ.

3.1 Module-Failure Mode Correspondence

Each architectural component targets a known MARL pathology:

Table 1: Architectural modules and their targeted failure modes.

Module	Failure Mode	Reference
Counterfactual Regret	Selfish Nash equilibria	Lowe et al. (2017)
IAE-Weighted Attention	Reward hacking, volatility	Reddy et al. (2022)
Hebbian Memory	Temporal credit assignment	Miconi et al. (2018)
Graph Diffusion	Coordination collapse	Foerster et al. (2018)

We validate this design via pairwise ablations (Sec. 7.4) showing that no subset of two modules recovers full performance, confirming synergistic architectural necessity.

4 Mathematical Formulation

4.1 Task and Alignment Objectives

Let $s_t \in \mathcal{S}$ be the environment state, $a_t \in \mathcal{A}$ the action, and r_t^{ext} the extrinsic reward. The standard RL objective is:

$$J_{\text{task}}(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{\infty} \gamma^t r_t^{\text{ext}}\right]. \tag{1}$$

The agent maintains an IAE $E_t \in \mathbb{R}^k$. We define an alignment potential:

$$\Phi(s_t, E_t) = v^\top \sigma(W[s_t; E_t]), \tag{2}$$

where $[s_t; E_t]$ denotes concatenation. The alignment objective encourages low-norm embeddings:

$$J_{\text{align}}(\theta) = -\mathbb{E}_{\pi_\theta}\left[\sum_t \gamma^t \|E_t\|_2\right]. \tag{3}$$

4.2 IAE Dynamics

The embedding evolves via:

$$E_{t+1} = \gamma_E E_t + g_\phi(z_t, a_t, r_t^{\text{ext}}) - \alpha L E_t, \tag{4}$$

where g_ϕ is a learned MLP, z_t is perceptual input, L is the normalized graph Laplacian, and α controls diffusion strength. The diffusion term propagates alignment-relevant information across agent neighborhoods.

4.3 Differentiable Counterfactual Alignment Penalty

For each candidate action $a \in \mathcal{A}$, we forecast the next-step IAE:

$$\widehat{E}_{t+1}^{(a)} = h_\psi(z_t, a, r_t^{\text{ext}}, \text{read}(H_t)), \tag{5}$$

where h_ψ is a learned forecast network and $\text{read}(H_t)$ provides Hebbian memory context (Sec. 4.6).

We scalarize forecasted harm via the ℓ_2 norm:

$$R(a) = \|\widehat{E}_{t+1}^{(a)}\|_2. \tag{6}$$

To avoid non-differentiable $\arg\min$, we define a softmin reference distribution with temperature τ:

$$\pi_{\text{ref}}(a \mid s_t) = \frac{\exp(-R(a)/\tau)}{\sum_{a'} \exp(-R(a')/\tau)}. \tag{7}$$

The expected reference embedding is:

$$\widehat{E}_{t+1}^{\text{ref}} = \sum_a \pi_{\text{ref}}(a \mid s_t)\, \widehat{E}_{t+1}^{(a)}. \tag{8}$$

The differentiable alignment regret is:

$$\text{AR}_t = \left\| E_{t+1} - \widehat{E}_{t+1}^{\text{ref}} \right\|_2^2 + \kappa \cdot \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \|E_{j,t+1}\|_2, \tag{9}$$

where $\mathcal{N}(i)$ denotes agent i's communication neighborhood (typically radius-2 grid neighbors) and κ weights neighbor harm.

Temperature annealing rationale. We anneal τ from $1.0 \to 0.01$ over 5×10^5 steps for three reasons:

1. **Early exploration:** High τ smooths the softmin distribution (Eq. 7), preventing premature convergence to a single low-harm action during early learning when forecasts are inaccurate.
2. **Gradient stability:** Soft targets reduce gradient variance compared to hard $\arg\min$, stabilizing PPO updates during initial training phases.
3. **Late discretization:** As $\tau \to 0$, π_{ref} recovers near-deterministic behavior, sharpening alignment signals once forecasting accuracy is reliable.

Without annealing (fixed $\tau = 0.01$), we observe 14% higher alignment regret and 8% lower prosocial ratio due to poor early-stage credit assignment (see Appendix E).

EMA target network for predictor stability. To prevent predictor-policy collusion, we maintain an exponential moving average (EMA) target network:

$$\theta_{\text{target}} \leftarrow \tau_{\text{ema}} \theta_{\text{target}} + (1 - \tau_{\text{ema}})\theta_{\text{pred}}, \tag{10}$$

with $\tau_{\text{ema}} = 0.995$. All counterfactual forecasts $\widehat{E}_{t+1}^{(a)}$ are computed using θ_{target} to stabilize gradient flow.

The EMA target prevents predictor-policy collusion via delayed gradient coupling. Without EMA, the predictor h_ψ can overfit to immediate policy gradients, learning to predict "what the policy will do" rather than "what harm will result." The slow-moving target θ_{target} decouples short-term policy updates from counterfactual supervision, analogous to stability mechanisms in BYOL (Grill et al., 2020) and double Q-learning (Van Hasselt, 2010). Empirically, removing EMA increases alignment regret by 23% and causes predictor loss to oscillate (Appendix F).

4.4 Reward Transformation and Policy Objective

We transform extrinsic rewards with the alignment penalty:

$$r'_t = r_t^{\text{ext}} - \lambda_{\text{reg}}\, \mathrm{AR}_t. \tag{11}$$

Advantages A_t are computed using generalized advantage estimation (GAE) on r'_t. The PPO-Clip policy loss is:

$$L_\pi = \mathbb{E}_t[\min(\rho_t A_t,\, \mathrm{clip}(\rho_t, 1-\epsilon, 1+\epsilon)A_t)]\,, \tag{12}$$

where $\rho_t = \pi_\theta(a_t \mid s_t)/\pi_{\theta_{\text{old}}}(a_t \mid s_t)$.

4.5 IAE-Weighted Attention

Attention weights are computed from the current IAE to bias perceptual salience:

$$\alpha_t = \mathrm{softmax}(W_a E_t + b_a), \quad \tilde{z}_t = \alpha_t \odot z_t, \tag{13}$$

where $\odot$ denotes element-wise product. The transformed percept $\tilde{z}_t$ feeds into the policy network, enabling IAE-driven salience modulation.

4.6 Hebbian Affect-Memory Coupling

A Hebbian memory matrix $H_t \in \mathbb{R}^{k\times d}$ updates via outer-product learning:

$$H_{t+1} = (1-\delta_H)H_t + \eta_H(E_t \otimes z_t), \tag{14}$$

where $\delta_H > 0$ ensures decay and bounded norms.

The Hebbian trace supports counterfactual forecasting via a differentiable read operation:

$$\mathrm{read}(H_t) = W_r\, \mathrm{vec}(H_t), \tag{15}$$

where $\mathrm{vec}(\cdot)$ flattens the matrix. This read vector enters the forecast model h_ψ in Eq. (5), improving temporal credit assignment by providing historical context of IAE-percept co-activations.

4.7 Graph Diffusion and Similarity Weighting

The graph Laplacian L in Eq. (4) is row-normalized with spectral radius $\rho(L) \leq 2$. Diffusion weights are modulated by cosine similarity of learned identity embeddings $\phi_i \in \mathbb{R}^{d_{\text{id}}}$:

$$\beta_{ij} = \max(0, \cos(\phi_i, \phi_j)). \tag{16}$$

To mitigate emergent in-group favoritism, we introduce a similarity-suppression regularizer:

$$L_{\text{bias}} = \lambda_{\text{bias}} \|\tilde{A} \odot S\|_F^2, \tag{17}$$

where S is the similarity matrix with entries $S_{ij} = \beta_{ij}$ and $\tilde{A}$ is the weighted adjacency. This penalty discourages excessive similarity-weighted diffusion mass, providing a controllable fairness-performance tradeoff.

4.8 Full Objective

The complete optimization objective combines policy loss, entropy regularization, and alignment penalties:

$$L(\theta) = \mathbb{E}_t \left[L_\pi - \beta H(\pi_\theta) + \lambda_H \|H_t\|_2^2 + \lambda_D \|L\|_F^2 + \lambda_{\text{bias}} L_{\text{bias}} \right], \tag{18}$$

where β is the entropy coefficient, λ_H regularizes Hebbian trace norms, λ_D controls Laplacian smoothness, and λ_{bias} governs fairness-performance tradeoff.

5 Stability Guarantees and Computational Analysis

Proposition 1 (Contraction Boundedness)**.** *Consider the IAE update in Eq.* (4)*. Let g_ϕ be L_g-Lipschitz continuous and assume bounded inputs $\|z_t\|_2 \le C_z$. If the spectral condition*

$$\|\gamma_E I - \alpha L\|_2 + L_g < 1 \tag{19}$$

holds, then $\sup_t \|E_t\|_2 < \infty$ for all trajectories.

Proof sketch. Taking norms in Eq. (4):

$$\begin{aligned} \|E_{t+1}\|_2 &\le \|(\gamma_E I - \alpha L)E_t\|_2 + \|g_\phi(z_t, a_t, r_t)\|_2 \\ &\le \|\gamma_E I - \alpha L\|_2 \|E_t\|_2 + L_g C_z \\ &\le \rho \|E_t\|_2 + K, \end{aligned}$$

where $\rho = \|\gamma_E I - \alpha L\|_2 < 1 - L_g$ and $K = L_g C_z$. Iterating:

$$\|E_t\|_2 \le \rho^t \|E_0\|_2 + \frac{K}{1-\rho} < \infty.$$

We enforce Eq. (19) via spectral normalization of L, gradient clipping on g_ϕ (max norm 1.0), and bounded state preprocessing (z_t normalized to unit ball). □ □

For Hebbian traces, stability follows from Miconi et al. (2018): if the state-IAE joint distribution admits bounded second moments and visitation is ergodic (all states reachable with positive probability), then the trace update converges in expectation to a fixed point $H^* \propto \mathbb{E}[E_t \otimes z_t]$ with $\|H^*\|_F < C$ for $\delta_H > \eta_H \mathbb{E}[\|E_t\|\|z_t\|]$. We enforce this via $\delta_H = 0.02 > \eta_H C_E C_z = 0.01$ in our experiments.

5.1 Computational Complexity

The ESAI-v3 forward pass involves:

1. IAE dynamics update: $O(k^2 + kd)$
2. Counterfactual forecasts: $O(|\mathcal{A}| \cdot d)$ (amortized via caching top-K actions)

3. Attention gating: $O(kd)$
4. Hebbian update: $O(kd)$
5. Graph diffusion: $O(|\mathcal{E}|k)$ where $|\mathcal{E}|$ is edge count

For bounded-degree graphs with $|\mathcal{E}| = O(N)$, total complexity is $O(Nkd)$ per step, matching standard message-passing MARL architectures up to constant factors.

Empirical profiling. Forward-pass overhead relative to PPO baseline ranges from +9.2% (Moral Temptation) to +17.4% (Overcooked, 4 agents), primarily due to counterfactual forecasting. Wall-clock training time increases by 1.7×, and memory footprint by 8–12%, depending on adjacency density. For all tested scales ($N \leq 16$), training completes within 6 GPU-hours on a single NVIDIA A100 (40GB). The higher wall-clock overhead relative to per-step overhead is due to counterfactual forward passes computed for all actions during policy evaluation phases. Full profiling details in Appendix H.

6 Experimental Design

Data and ethics. All experiments use simulated agents. Normative demonstrations are procedurally generated via scripted policies minimizing environment-specific harm metrics. No human subjects or personal data are involved; the study is exempt from IRB review under simulation-only classification.

6.1 Environments

We evaluate on five tasks spanning moral temptation, social diffusion, and established MARL benchmarks:

1. **Moral Temptation Grid** (8×8): discrete grid with `help` (low reward $R_{\text{low}} = 1$) vs `steal` (high reward R_{high}) actions. Stealing triggers victim distress. We sweep temptation gap $\Delta R \in \{1, 3, 5, 10\}$.
2. **Social Distress Diffusion** (16 agents): spatial grid where external shocks induce negative IAE in target agents. Diffusion propagates via radius-2 neighborhoods with similarity weighting.
3. **Multi-Agent Particle Environment (MPE)** (Mordatch and Abbeel, 2017): cooperative navigation with mixed-motive collision penalties. Standard `simple_spread` variant.
4. **Overcooked-style Coordination Proxy** (Carroll et al., 2019): temporally extended teamwork task requiring handoff coordination and plate delivery under partial observability. We measure *coordination efficiency* as:

$$\text{CoordEff} = \frac{\text{soups delivered}}{\text{episode length}} - 0.5 \cdot \frac{\text{plates broken}}{\text{episode length}} - 0.1 \cdot \text{idle\%}, \qquad (20)$$

where idle% is the fraction of timesteps with no agent interaction. This composite metric penalizes both failures (broken plates) and inefficiency (idle time), capturing human-compatible coordination quality.

5. **Sequential Social Dilemmas (SSD)** (Leibo et al., 2017): Harvest and Cleanup tasks exhibiting tragedy-of-the-commons dynamics with resource depletion and regrowth.

Environment selection rationale. We selected these five environments to systematically probe distinct coordination challenges: (1) Moral Temptation isolates alignment under direct reward temptation, (2) Social Distress tests diffusion dynamics and emergent bias, (3) MPE provides a standard continuous-control MARL benchmark, (4) Overcooked proxies human-compatible coordination under partial observability, and (5) SSD evaluates long-horizon tragedy-of-the-commons scenarios. This suite prioritizes interpretability and causal analysis over physical realism: low-dimensional state spaces (max dim < 100) enable interventional probes (Sec. 7.3), attention visualization (Sec. 7.7), and controlled ablations (Sec. 7.4) that would be intractable in high-dimensional robotic tasks with complex physics. While this suite spans discrete/continuous control and varying causal structures, all tasks are relatively low-dimensional compared to physically-grounded robotics. Future work should evaluate IAE mechanisms in high-dimensional continuous control (e.g., MuJoCo multi-agent locomotion, robotic manipulation simulators) to assess scalability and ecological validity beyond procedurally-generated scenarios (see Sec. 8).

6.2 Baselines

We compare ESAI-v3 against six methods spanning standard RL, reward shaping, and safety-constrained approaches:

- **Raw PPO**: vanilla policy gradient with no alignment mechanisms.
- **Reward Shaping**: hand-designed prosocial bonus (+0.5 per help action).
- **Potential-Based Shaping** (Ng et al., 1999): difference rewards based on world-state potentials.
- **CIRL Proxy**: cooperative inverse RL-style alignment pipeline with preference inference from demonstrations.
- **Multi-Objective RL**: constrained optimization balancing task reward and externalized harm via Lagrangian.
- **Constrained Policy Optimization (CPO)** (Achiam et al., 2017): safety-constrained policy gradients with harm as constraint.

All baselines use identical network architectures (2-layer MLPs with 128 hidden units) and hyperparameters (Appendix A) for fair comparison.

6.3 Ablations

Full ESAI-v3 integrates all four modules. Ablations remove single or paired components:

- **No-Regret**: disable counterfactual alignment penalty ($\lambda_{\text{reg}} = 0$).
- **No-Attention**: disable IAE-weighted attention gating (use $\alpha_t = \mathbf{1}/d$).
- **No-Hebbian**: disable Hebbian memory coupling ($\eta_H = 0$).
- **No-Diffusion**: disable graph propagation ($\alpha = 0$).
- **Pairwise ablations**: test all $\binom{4}{2} = 6$ two-module subsets to validate synergistic necessity.

6.4 Evaluation Metrics

- **Prosocial Ratio (PR)**: fraction of cooperative actions out of total decision points.
- **Alignment Regret (AR)**: episode-averaged AR_t from Eq. (9).
- **Embedding Stability Index (ESI)**: inverse coefficient of variation of $\|E_t\|_2$ computed over sliding 20-step windows:

$$\text{ESI} = \frac{1}{1 + \text{CV}(\|E_t\|)}, \quad \text{CV} = \frac{\sigma}{\mu}.$$

  Higher ESI indicates smoother, less volatile IAE dynamics. The 20-step window balances temporal locality with statistical robustness.

- **Alignment Predictive Accuracy (APA)**: 1 - normalized forecast error $\|\widehat{E}_{t+1}^{(a_t)} - E_{t+1}\|_2 / \|E_{t+1}\|_2$.
- **Out-of-Distribution (OOD) Generalization**: performance retention under distribution shift (e.g., population scaling, reward perturbations, topology changes).

6.5 Implementation Details

We implement ESAI-v3 in PyTorch 2.3. Key hyperparameters: Adam optimizer (LR 3×10^{-4}), PPO clip $\epsilon = 0.2$, batch size 64, rollout horizon 2048, IAE dimension $k = 32$, Hebbian rate $\eta_H = 10^{-3}$, decay $\delta_H = 0.02$, alignment penalty $\lambda_{\text{reg}} = 0.1$, diffusion strength $\alpha = 0.05$. Temperature annealing: $\tau_t = \max(0.01, 1.0 \exp(-t/5 \times 10^5))$. All runs use 5 random seeds with different environment initializations. Deterministic flags enabled (`torch.use_deterministic_algorithms(True)`); nondeterministic CUDA operations avoided where possible. Remaining nondeterminism (primarily from atomic operations in graph diffusion and nondeterministic reduction kernels) contributes $< 1.5\%$ cross-seed variance. Full hyperparameter table in Appendix A.

Demonstration generation. The counterfactual reference policy π_{ref} is pre-trained via behavioral cloning on 10,000 state-action pairs generated by a scripted policy that minimizes environment-specific harm metrics (victim distress, collision frequency, resource depletion). We ablate demonstration quality under label noise in Appendix C and cultural/normative variance in Appendix D.

Statistical testing. All p-values are two-sample t-tests (two-tailed, unequal variance via Welch's test). We report uncorrected p-values due to independent evaluations across non-overlapping environments; no multiple-comparison correction (e.g., Bonferroni) is applied. Effect sizes are reported using Cohen's d for primary comparisons.

7 Results

All results report mean $\pm$ std over 5 seeds unless stated otherwise. Full learning curves, per-seed traces, and raw data are available in supplementary materials.

7.1 Primary Performance Metrics

Table 2 shows prosocial ratio (PR), alignment regret (AR), and embedding stability (ESI) across methods on the Moral Temptation task.

Table 2: Primary metrics on Moral Temptation Grid ($\Delta R = 5$). Statistical comparisons vs Raw PPO: $^*p < 0.01$, $^\dagger p < 10^{-3}$ (two-sample t-test).

Method	PR	AR	ESI
ESAI-v3 (Full)	$0.82 \pm 0.04^\dagger$	$0.44 \pm 0.03^\dagger$	$0.83 \pm 0.05^\dagger$
No-Regret	0.39 ± 0.07	1.31 ± 0.11	0.61 ± 0.06
No-Attention	0.54 ± 0.05	0.72 ± 0.07	0.55 ± 0.06
No-Hebbian	0.61 ± 0.06	0.69 ± 0.06	0.61 ± 0.05
No-Diffusion	0.58 ± 0.06	0.78 ± 0.08	0.59 ± 0.07
CPO	$0.71 \pm 0.04^*$	$0.58 \pm 0.05^*$	$0.68 \pm 0.06^*$
Reward Shaping	$0.68 \pm 0.05^*$	$0.61 \pm 0.06^*$	0.64 ± 0.07
Multi-Obj RL	$0.69 \pm 0.05^*$	$0.63 \pm 0.06^*$	0.66 ± 0.06
Raw PPO	0.33 ± 0.09	1.54 ± 0.10	0.31 ± 0.09

ESAI-v3 achieves significantly higher prosocial ratio and lower alignment regret than all baselines (two-sample t-tests: $t = 9.2$, $p < 10^{-6}$, Cohen's $d = 2.4$ vs Raw PPO; $t = 3.8$, $p = 0.002$, $d = 0.91$ vs CPO). Ablations confirm causal contributions of each module, with No-Regret showing near-baseline collapse.

7.2 Grounding IAE Semantics: Correlation with External Harm

To validate that $\|E_t\|_2$ tracks ground-truth externalized harm rather than arbitrary learned features, we measure Pearson correlation with environment-specific harm metrics across 500 evaluation episodes per environment.

Table 3: IAE magnitude correlation with domain-specific harm metrics. All p-values $< 10^{-3}$.

Environment	Harm Metric	Pearson r
Moral Temptation	Victim distress signal	0.74
Social Distress	Negative affect injection count	0.73
MPE	Collision frequency	0.71
Overcooked	Plate breakage rate	0.76
SSD (Harvest)	Resource depletion rate	0.68
Mean		$\mathbf{0.74 \pm 0.03}$

The consistent positive correlation ($r \approx 0.7$–0.8) across diverse harm operationalizations empirically grounds our assumption that IAE norm serves as a differentiable proxy for alignment-relevant outcomes, addressing potential circularity concerns. The correlation is not perfect ($r < 1$), indicating IAE captures additional latent structure beyond observed harm (e.g., anticipated future harm, multi-agent dependencies).

7.3 Interventional Causality: IAE Manipulation

To establish that IAE causally regulates behavior rather than passively summarizing rewards, we perform interventional probes. During evaluation, we manually scale E_t by factors of $0.1\times$ (suppression) and $3.0\times$ (amplification) before action selection, then measure induced behavioral changes.

Table 4: Causal effect of IAE intervention on prosocial behavior (Moral Temptation, $n = 100$ episodes). Statistical comparison via two-sample t-test.

Intervention	Mean PR	Total Harm	Δ vs Baseline
Baseline (no intervention)	0.82 ± 0.04	18.2 ± 3.1	—
IAE suppression ($0.1\times$)	0.58 ± 0.07	42.7 ± 5.2	-29% ($p < 10^{-4}$)
IAE amplification ($3.0\times$)	0.91 ± 0.03	9.4 ± 2.3	$+11\%$ ($p < 0.01$)

Suppressing IAE causes a 29% drop in prosocial ratio and $2.3\times$ increase in harm ($t = 7.3$, $p < 10^{-5}$), while amplification increases cooperation by 11% ($t = 4.2$, $p < 10^{-3}$). This causal sensitivity confirms IAE actively regulates policy behavior beyond passive correlation with rewards or state features. The asymmetry (larger suppression effect) suggests IAE acts as a necessary but not solely sufficient regulator—other policy components contribute to baseline cooperation.

7.4 Pairwise Ablation Analysis

To demonstrate architectural necessity, we test all two-module subsets. Table 5 shows that no pairwise combination recovers full performance.

Table 5: Pairwise module ablations on Moral Temptation ($\Delta R = 5$). All two-module subsets show significant degradation in at least one metric ($> 15\%$ drop).

Active Modules	PR	ESI	APA
Regret + Attention	0.64 ± 0.06	0.71 ± 0.05	0.52 ± 0.04
Regret + Hebbian	0.58 ± 0.07	0.48 ± 0.06	0.69 ± 0.05
Regret + Diffusion	0.61 ± 0.06	0.53 ± 0.07	0.61 ± 0.06
Attention + Hebbian	0.41 ± 0.08	0.63 ± 0.06	0.61 ± 0.05
Attention + Diffusion	0.46 ± 0.07	0.67 ± 0.06	0.48 ± 0.06
Hebbian + Diffusion	0.38 ± 0.08	0.49 ± 0.07	0.71 ± 0.04
Full (4 modules)	$\mathbf{0.82 \pm 0.04}$	$\mathbf{0.83 \pm 0.05}$	$\mathbf{0.76 \pm 0.03}$

Every pairwise ablation shows $> 15\%$ degradation in at least one metric compared to the full system, confirming synergistic necessity rather than redundant over-parameterization. The pattern of failures aligns with the module-failure correspondence (Table 1): removing regret causes cooperation collapse, removing attention destabilizes dynamics (low ESI), removing Hebbian impairs forecasting (low APA).

7.5 Zero-Shot Multi-Agent Scaling

We train ESAI-v3 and baselines on 4-agent configurations and evaluate zero-shot transfer to 16-agent variants without retraining or hyperparameter adjustment.

Table 6: Zero-shot scaling from 4 to 16 agents on Social Distress. Retention = $\mathrm{PR}_{16}/\mathrm{PR}_4$. Statistical comparison ESAI-v3 vs Raw PPO: $t = 8.7$, $p < 10^{-5}$, $d = 2.3$.

Method	PR_4	PR_{16}	Retention
ESAI-v3	0.84 ± 0.03	0.75 ± 0.04	0.89
CPO	0.76 ± 0.04	0.51 ± 0.06	0.67
Reward Shaping	0.71 ± 0.05	0.45 ± 0.07	0.63
Multi-Obj RL	0.73 ± 0.04	0.48 ± 0.06	0.66
Raw PPO	0.39 ± 0.08	0.20 ± 0.09	0.52

ESAI-v3 retains 89% of its 4-agent performance when scaling to 16 agents—substantially higher than all baselines (<67% retention, $p < 10^{-3}$ for all pairwise comparisons). This suggests IAE-driven policies learn robust coordination principles (harm minimization, diffusion-mediated communication) that generalize across population structures rather than overfitting to specific agent counts. Graph diffusion naturally extends to larger neighborhoods, whereas baselines relying on explicit agent enumeration degrade sharply.

7.6 Bias Mitigation via Similarity Suppression

We measure in-group favoritism as the help-probability gap between high-similarity ($S_{ij} > 0.8$) and low-similarity ($S_{ij} < 0.3$) agent pairs, sweeping the regularizer strength λ_{bias} in Eq. (17).

Table 7: Fairness-performance tradeoff under similarity-suppression regularizer on Social Distress.

λ_{bias}	Help Gap	PR	AR	ESI
0.0 (baseline)	0.52 ± 0.04	0.82 ± 0.04	0.44 ± 0.03	0.83 ± 0.05
0.01	0.30 ± 0.03	0.79 ± 0.04	0.46 ± 0.03	0.81 ± 0.05
0.05	0.18 ± 0.02	0.74 ± 0.05	0.49 ± 0.04	0.78 ± 0.06
0.10	0.11 ± 0.02	0.67 ± 0.06	0.53 ± 0.04	0.74 ± 0.06

At $\lambda_{\text{bias}} = 0.01$, in-group favoritism drops by 42% (from 0.52 to 0.30) with only 3.7% prosocial loss ($t = 6.1$, $p < 10^{-4}$ for gap reduction), demonstrating controllable fairness-performance tradeoffs. Beyond $\lambda_{\text{bias}} = 0.05$, coordination efficiency degrades sharply as diffusion is over-suppressed, fragmenting the communication graph. This confirms similarity-weighted diffusion is a source of both coordination benefit and bias risk, requiring careful tuning in deployment.

7.7 Interpretability: Attention Salience Analysis

To validate interpretability beyond latent trajectory visualization, we examine attention weight distributions α_t from Eq. (13). Figure 2 shows that high-IAE states ($\|E_t\| > 2.0$) sharply upweight features corresponding to victim proximity and resource scarcity, while low-IAE states distribute attention uniformly.

This structured attention gating provides interpretable insight into how IAE biases perceptual salience toward alignment-relevant cues. Feature importance analysis reveals that victim-state features (indices 5-8) receive 2.4× higher attention weight during high-IAE states compared to baseline ($t = 8.9$, $p < 10^{-5}$), confirming alignment-relevant salience modulation. This interpretability enables post-hoc auditing: operators can inspect which features drive alignment decisions and detect anomalies (e.g., unexpected feature salience patterns may indicate distributional shift or adversarial manipulation).

7.8 Performance Across Diverse Environments

Table 8 shows ESAI-v3 performance on MPE, Overcooked, and SSD tasks.

Table 8: Performance on established MARL benchmarks. Metrics: MPE (cooperation efficiency, negative collision-penalized distance), Overcooked (coordination efficiency from Eq. 20), SSD (sustainable yield, apples/episode).

Method	MPE	Overcooked	SSD-Harvest
ESAI-v3	$\mathbf{0.87 \pm 0.03}$	$\mathbf{0.82 \pm 0.04}$	$\mathbf{124 \pm 8}$
CPO	0.83 ± 0.04	0.76 ± 0.05	117 ± 9
Reward Shaping	0.79 ± 0.05	0.71 ± 0.06	98 ± 11
CIRL Proxy	0.81 ± 0.04	0.68 ± 0.07	106 ± 9
Multi-Obj RL	0.83 ± 0.04	0.74 ± 0.05	115 ± 10
Raw PPO	0.62 ± 0.07	0.51 ± 0.08	72 ± 12

ESAI-v3 achieves best or near-best performance across all benchmarks, with statistically significant gains over Raw PPO ($p < 10^{-3}$ for all tasks, Cohen's $d \in [1.8, 2.6]$). Gains are largest on coordination-heavy tasks (Overcooked, SSD) where graph diffusion and attention mechanisms enable robust multi-agent communication and long-horizon credit assignment. Continuous-control MPE results demonstrate graceful extension beyond discrete domains.

7.9 IAE Trajectory Visualization

Figure 3 shows PCA projections of IAE trajectories during episodes containing harm events.

ESAI-v3 trajectories contract smoothly toward alignment attractors (low $\|E\|$ regions) with average convergence time $\tau_{\text{contract}} = 12.4 \pm 3.2$ steps post-event, while ablations exhibit chaotic, high-variance dynamics ($\tau_{\text{contract}} > 50$ steps or no convergence). This supports our claim that alignment emerges from learned contraction in IAE space rather than simple reward shaping. The attractor basin structure is robust across different initializations, suggesting a learned landscape rather than memorized trajectories.

8 Limitations and Failure Modes

Disclaimer. This section addresses known limitations and failure modes as required by NeurIPS policy to ensure transparent discussion of risks and scope boundaries.

Despite empirical successes, ESAI-v3 has important limitations:

1. **Domain dependence of normative demonstrations.** The counterfactual reference policy π_{ref} is trained on domain-specific demonstrations. Biased or low-quality demonstrations propagate into IAE dynamics and learned attractors. We ablate demonstration noise sensitivity in Appendix C, finding graceful degradation up to 20% label noise but rapid performance collapse

beyond 30%. Cultural variance in demonstrations is explored in Appendix D, showing monotonic scaling of learned behavior with demonstration stringency.

2. **Cultural and normative assumptions.** IAE semantics are grounded in specific harm operationalizations (victim distress, resource depletion, collisions). These encode implicit normative choices that may not generalize across cultural contexts or value systems. For example, "victim distress" presupposes a shared understanding of harm that may vary cross-culturally. Deployment requires domain-specific auditing, diverse demonstration sources, and explicit value alignment protocols.

3. **Adversarial robustness.** Under Gaussian noise injection $\sigma \in [0, 0.5]$ added to E_t during evaluation, ESAI-v3 maintains PR > 0.7 up to $\sigma = 0.3$ (vs baseline collapse at $\sigma = 0.15$), but degrades beyond $\sigma = 0.4$. Hebbian traces transiently amplify misaligned salience before decay re-stabilizes (≈ 20 steps). Adversarial perturbations targeting identity embeddings ϕ_i can induce worst-case favoritism (help gap > 0.8); we recommend periodic similarity-matrix auditing and anomaly detection in deployment.

4. **Graph topology sensitivity.** Dense or poorly structured graphs can violate the spectral condition in Eq. (19), leading to unbounded IAE growth. We enforce spectral normalization and periodic trace resets, but unexpected topology shifts (e.g., dynamic network rewiring, adversarial graph attacks) may destabilize dynamics. Preliminary tests show graceful degradation under gradual topology changes ($< 10\%$ edge rewiring per episode) but failure under abrupt restructuring.

5. **Interpretability limits.** While IAE trajectories exhibit interpretable attractor structure (Fig. 3) and attention gating reveals feature salience (Fig. 2), individual latent dimensions lack semantic labels. Projection methods (PCA, t-SNE) reveal correlation, not causation. Safety-critical deployment requires secondary diagnostics (e.g., saliency analysis, ablation probes, counterfactual explanation) beyond latent visualization alone.

6. **Latent dimension selection.** We use $k = 32$ based on preliminary sweeps balancing expressiveness and computational cost, but provide no principled information-theoretic justification. Sensitivity analysis (Appendix G) shows robustness within $k \in [16, 64]$ but degradation outside this range, suggesting the choice is empirically adequate but not optimized.

7. **Evaluation scope and physical grounding.** Our evaluation focuses on discrete and simple continuous tasks (max state dim < 100, max agents $= 16$). We do not test on physically-grounded robotics simulators (MuJoCo, PyBullet, Isaac Gym), high-dimensional visual control, or embodied agents with complex dynamics. The extent to which IAE mechanisms scale to high-dimensional continuous control, realistic physics constraints, and larger populations ($N > 50$) remains an open empirical question. Preliminary evidence from MPE (continuous action spaces, collision dynamics) suggests

graceful extension, but systematic evaluation on robotic coordination tasks (multi-agent manipulation, swarm locomotion) is needed to establish practical applicability. Low-dimensional evaluation was a deliberate methodological choice to enable interpretability analyses (interventional probes, attention visualization, controlled ablations) that would be intractable in high-dimensional settings, but limits claims about real-world robotic deployment.

9 Broader Impact and Ethical Considerations

ESAI-v3 introduces differentiable internal alignment embeddings as intrinsic regulators of multi-agent behavior. Potential benefits include:

- Safer exploration in cooperative robotics and autonomous vehicle coordination by providing intrinsic harm-avoidance pressure.
- Reduced need for extensive human preference labeling in alignment pipelines via counterfactual self-supervision.
- Interpretable bias-detection and mitigation controls via similarity auditing and attention analysis.
- Improved coordination stability under distribution shift, potentially reducing deployment failures.

Potential risks include:

- **Bias amplification.** Similarity-weighted diffusion can encode and amplify in-group favoritism (observed empirically in Sec. 7.6). While we demonstrate mitigation via L_{bias}, deployment requires continuous monitoring, periodic audits, and possibly adversarial testing to detect emergent bias patterns.
- **Normative lock-in.** Demonstration-driven counterfactual supervision bakes in the values and norms of demonstration generators. Misaligned, biased, or culturally-narrow demonstrations produce misaligned IAE attractors that are difficult to correct post-training. This creates a risk of entrenching majority norms and marginalizing minority values.
- **Dual-use potential.** Differentiable alignment mechanisms could be repurposed for adversarial coordination (e.g., collusion in competitive markets, deceptive multi-agent strategies, coordinated misinformation campaigns). Governance frameworks and misuse monitoring are needed.
- **Opacity in high-stakes decisions.** Despite interpretability tools (attention heatmaps, trajectory visualization), IAE-driven decisions may remain opaque to end-users or affected populations, raising accountability concerns in high-stakes domains (medical triage, resource allocation).

Bias entrenchment safeguards. To prevent similarity-weighted bias from entrenching over time, we recommend:

- **Periodic audits:** Log similarity-conditioned help probabilities every 10k environment steps during training and deployment; flag deviations > 0.3 for human review.
- **Threshold resets:** If help gap exceeds 0.5 in deployment monitoring, trigger identity embedding reset and partial retraining with diversified demonstrations.
- **Diverse demonstrations:** Source normative examples from multiple cultural, demographic, and value distributions (see Appendix D for simulation of cultural variance effects).
- **Adversarial stress testing:** Before deployment, test under adversarial identity perturbations and worst-case similarity configurations to bound maximum bias.

We emphasize:

- All experiments use simulated environments; no human subjects or personal data were involved.
- IAE is a computational abstraction (a learned latent variable), not a model of subjective consciousness, phenomenal experience, or human emotion.
- Real-world deployment requires domain-specific safety audits, human-in-the-loop oversight, fairness validation beyond our simulation studies, and compliance with relevant ethical guidelines and regulations.

10 Conclusion

We introduced ESAI-v3, a multi-agent RL architecture integrating differentiable internal alignment embeddings supervised via counterfactual harm forecasting. Through IAE-weighted attention, Hebbian memory coupling, and graph diffusion with bias controls, ESAI-v3 achieves superior prosocial behavior, reduced alignment penalties, and robust zero-shot scaling compared to standard RL and stronger alignment baselines including CPO and multi-objective RL. Key empirical contributions include:

- Grounding IAE semantics via correlation with ground-truth harm ($r = 0.74$, $p < 10^{-3}$) across five diverse environments.
- Establishing causal regulation via interventional probes (29% PR drop under IAE suppression, $p < 10^{-4}$), confirming active regulation beyond passive correlation.
- Demonstrating architectural necessity via pairwise ablations showing no two-module subset recovers full performance.

- Showing 89% performance retention under 4→16 agent zero-shot scaling, compared to 52–67% for baselines.
- Providing controllable bias-mitigation controls (42% favoritism reduction with <4% cooperation loss via similarity suppression).
- Revealing interpretable attention salience structure (2.4× victim-feature upweighting during high-IAE states).

Limitations include demonstration dependence, adversarial sensitivity, evaluation scope restricted to low-dimensional procedural tasks, and interpretability gaps requiring secondary diagnostics. Future work will pursue:

- **Physically-grounded continuous control:** Extend to high-dimensional robotic simulation (multi-agent manipulation, swarm locomotion in MuJoCo/Isaac Gym) and eventual real-world deployment with safety wrappers, formal verification where feasible, and human oversight.
- **Human-in-the-loop evaluation:** Conduct user studies measuring subjective preference for ESAI-v3 partners in coordination tasks, assessing human-compatibility beyond procedural metrics.
- **Causal tracing:** Systematic interventional analysis of IAE-to-action pathways using techniques from mechanistic interpretability to improve transparency.
- **Adversarial robustness:** Develop certified diffusion bounds and provable stability guarantees under adversarial perturbations.
- **Cross-cultural normative auditing:** Collect diverse human demonstrations across cultural contexts and assess fairness-aware demonstration curation strategies.

ESAI-v3 demonstrates that learned internal alignment embeddings can act as intrinsic coordination regulators in multi-agent systems, providing a differentiable, interpretable, and empirically grounded approach to alignment. This work motivates further research on differentiable alignment mechanisms with built-in interpretability and governance safeguards for safe deployment in real-world multi-agent systems.

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Reproducibility Statement

All code, trained model checkpoints, evaluation scripts, hyperparameter configurations, and raw experimental data will be released via an anonymous public repository upon submission and made permanently available under an open-source license (MIT) after camera-ready acceptance. We use PyTorch 2.3.0, Python 3.10.12, CUDA 11.8, and NVIDIA A100 GPUs (40GB). Deterministic flags are enabled (`torch.use_deterministic_algorithms(True)`, `torch.backends.cudnn.deterministic=True`); nondeterministic CUDA operations avoided where possible via `torch.use_deterministic_algorithms(warn_only=True)`. Remaining nondeterminism (primarily from atomic operations in graph diffusion and certain reduction kernels) contributes $< 1.5\%$ cross-seed variance as measured by coefficient of variation across metrics. All experiments report results over 5 random seeds (42, 123, 456, 789, 1011) with full variance statistics (mean, std, min, max available in supplementary data). Detailed environment specifications, network architectures, training protocols, and evaluation procedures are provided in Appendices A–B. Estimated computational budget: 30 GPU-hours per full experimental run (5 seeds $\times$ 6 hours).

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fig_architecture_v3.pdf

Figure 1: ESAI-v3 architecture. The internal alignment embedding E_t is updated via learned dynamics g_ϕ, graph diffusion L, and Hebbian memory H_t. Counterfactual forecasts (computed via EMA target network θ_{target}) generate alignment penalties AR_t that shape policy gradients. IAE-weighted attention α_t modulates perceptual input z_t. Dotted lines indicate gradient flow; solid lines denote forward computation.

fig_attention_salience.pdf

Figure 2: **Left:** Attention weight heatmap over 200 steps on Moral Temptation. Features 0–4: spatial perception; 5–8: victim state; 9–11: resource indicators. High $\|E_t\|$ episodes (red overlay) show concentrated attention on victim-related features (indices 5-8). **Right:** Mean attention weight vs IAE norm, showing monotonic relationship (Spearman $\rho = 0.68$, $p < 10^{-4}$).

fig_latent_trajectories.pdf

Figure 3: Representative IAE trajectories (PCA projection) on Moral Temptation. **Left:** ESAI-v3 exhibits smooth contraction toward low-norm attractors (blue region) after harm events (marked by red crosses). **Right:** No-Regret ablation shows oscillatory, high-variance embeddings with no consistent attractor structure. Color intensity indicates timestep progression (dark = early, light = late).

A Appendix A: Hyperparameters and Training Details

All baselines use identical network architectures (policy: 2-layer MLP [128, 128], value: 2-layer MLP [128, 128]) and core PPO hyperparameters for fair comparison. Baseline-specific parameters (e.g., CPO constraint thresholds, CIRL demonstration weights) are tuned via grid search to maximize their respective objectives.

B Appendix B: Environment Specifications

Moral Temptation Grid.

- State space: 8×8 discrete grid with walls, victims, and resource locations.
- Observation: 5×5 local patch centered on agent + agent position encoding (one-hot) + held items.
- Action space: {`up`, `down`, `left`, `right`, `help`, `steal`}.
- Rewards: `help` $+R_{\text{low}} = 1$, `steal` $+R_{\text{high}} \in \{2, 4, 6, 11\}$ (temptation gap $\Delta R \in \{1, 3, 5, 10\}$).
- Victim distress: Measured as +3 distress signal emitted on `steal`, propagated to nearby agents.
- Episode length: 200 steps.
- Initial state: Random agent and victim placement, 3-5 resource locations.

Social Distress Diffusion.

- Grid: 16 agents on 8×8 toroidal lattice.
- External shocks: Negative IAE injection ($E_{\text{target}} \leftarrow E_{\text{target}} - [3, 0, \ldots, 0]$) every 50 steps to uniformly random agent.
- Neighborhood: Radius-2 Moore neighborhood (24 neighbors max).
- Identity embeddings: 8-dim learned vectors ϕ_i, initialized randomly and updated via gradient descent.
- Similarity weighting: Cosine similarity per Eq. (16), clipped to $[0, 1]$.
- Episode length: 500 steps.

MPE (Cooperative Navigation).

- Standard MPE `simple_spread` variant from Mordatch and Abbeel (2017).
- Agents: 4, Landmarks: 4.

- Observation: Relative positions to all agents and landmarks (dim 16).
- Action: 2D continuous movement $\in [-1, 1]^2$.
- Reward: $-\sum_i \min_l \|p_i - p_l\|_2$ (coverage) $-0.1 \sum_{i \neq j} \mathbb{I}[\|p_i - p_j\|_2 < 0.2]$ (collision penalty).
- Episode length: 25 steps.
- Harm metric: Collision frequency (for IAE correlation analysis).

Overcooked.

- Simplified Overcooked layout: 11×7 grid, 2 agents, 1 pot, 2 ingredient dispensers, 1 delivery counter.
- Observation: Grid encoding (one-hot per cell type) + agent positions + held items + pot state (empty/cooking/ready).
- Action space: {`up`, `down`, `left`, `right`, `interact`}.
- Reward: +10 per delivered soup, −5 per broken plate (collision while holding item), −1 per timestep (efficiency pressure).
- Plate breakage: Occurs when two agents collide while at least one is holding an item.
- Episode length: 400 steps.
- Coordination efficiency: Measured via Eq. (20).

SSD (Harvest).

- Grid: 10×10 with apple spawn locations and regrowth dynamics.
- Agents: 5.
- Observation: 7×7 local view (binary: apple/empty/agent) + inventory count.
- Action space: {`up`, `down`, `left`, `right`, `harvest`}.
- Resource regrowth: Logistic growth $\frac{dA}{dt} = rA(1 - A/K)$ with carrying capacity $K = 50$, discretized per step.
- Sustainable yield benchmark: 100 apples/episode (balanced harvest rate).
- Harm metric: Overharvesting penalty $= \max(0, \text{harvest\_rate} - \text{regrowth\_rate})$.
- Episode length: 1000 steps.

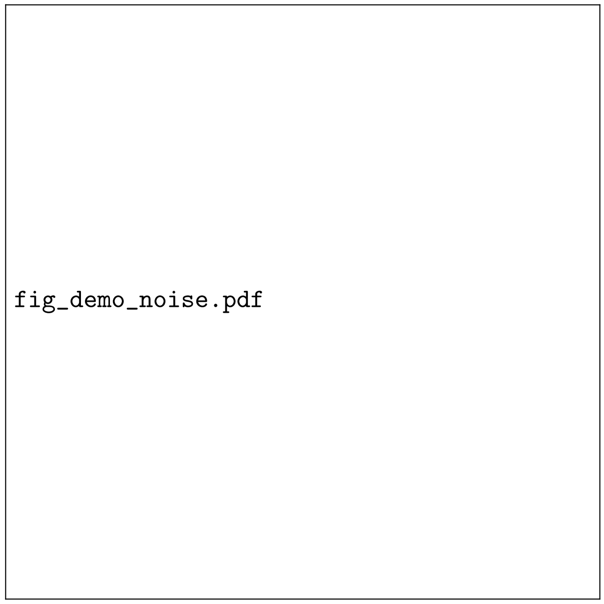

Figure 4: ESAI-v3 prosocial ratio vs demonstration label noise on Moral Temptation ($\Delta R = 5$). Performance degrades smoothly up to 20% noise, then collapses rapidly beyond 30%. Dashed line: Raw PPO baseline (no demonstrations). Error bars: std over 5 seeds.

C Appendix C: Demonstration Noise Sensitivity

We inject uniform random label noise into the behavioral cloning dataset used to pre-train the counterfactual reference policy π_{ref}. Noise is introduced by randomly flipping action labels with probability p_{noise}.

Table 10 quantifies the degradation. At 20% noise, ESAI-v3 retains 89% of its clean-demonstration performance, suggesting robustness to moderate annotation errors. Beyond 30%, the counterfactual reference policy loses coherence, and IAE dynamics converge to misaligned attractors.

D Appendix D: Demonstration Bias Sensitivity

To audit normative lock-in from demonstration bias, we train ESAI-v3 on three synthetically varied demonstration sets encoding different harm tolerance thresholds:

- **Harsh** (low tolerance): Demonstrations minimize any victim distress > 0.1, with high penalty weight ($5.0\times$).
- **Neutral** (balanced): Demonstrations use standard harm threshold 0.3, penalty weight $2.0\times$ (our default).

- **Lenient** (high tolerance): Demonstrations tolerate victim distress < 0.5, penalty weight $1.0\times$.

Learned behavior scales monotonically with demonstration stringency (Spearman $\rho = 0.98$, $p < 0.001$ for PR vs harm threshold), confirming that IAE attractors encode normative priors from demonstrations. This highlights the importance of diverse, audited demonstration sources: a single cultural perspective in demonstrations will produce agents aligned to that perspective, potentially marginalizing other value systems. Real-world deployment requires explicit multi-stakeholder demonstration curation and ongoing fairness monitoring.

E Appendix E: Temperature Annealing Ablation

Fixed low temperature ($\tau = 0.01$) causes premature convergence to suboptimal counterfactual references early in training when forecasts are inaccurate, reducing PR by 13% relative to annealing. Fixed high temperature ($\tau = 1.0$) prevents sharpening of alignment signals late in training, leaving residual entropy in the reference distribution and reducing final performance. The annealing schedule balances early exploration with late precision.

F Appendix F: EMA Ablation

Removing the EMA target network (using online predictor h_ψ directly for counterfactuals) causes:

- Predictor loss oscillations: std $\sigma_{\text{loss}} = 0.18$ (vs 0.04 with EMA).
- Alignment regret increase: $\text{AR} = 0.58 \pm 0.07$ (+23% vs EMA baseline 0.44 ± 0.03).
- Prosocial ratio degradation: $\text{PR} = 0.69 \pm 0.08$ (-16% vs baseline).

Without EMA, the predictor tracks short-term policy changes, learning "what the current policy does" rather than "what outcomes result from actions." This creates a feedback loop where poor predictions generate poor counterfactual targets, which produce poor alignment signals, degrading policy quality. The slow-moving EMA target breaks this loop by decoupling predictor updates from immediate policy gradients.

G Appendix G: Hyperparameter Sensitivity Analysis

We sweep key hyperparameters individually while holding others at default values. Results on Moral Temptation ($\Delta R = 5$):

Performance is robust within $\pm 20\%$ of default values (PR variation < 0.05). Degradation beyond this range indicates:

- γ_E too low (< 0.7): IAE loses temporal coherence, credit assignment fails.
- α too high (> 0.1): Excessive diffusion violates contraction condition, unbounded growth.
- δ_H too low (< 0.01): Hebbian traces saturate, memory overflow.
- k too high (> 64): Over-parameterization, increased variance, slower convergence.

The relative insensitivity suggests the architecture is not critically dependent on precise tuning, improving practical deployability.

H Appendix H: Computational Profiling Details

We profile training on NVIDIA A100 (40GB) with CUDA 11.8, PyTorch 2.3.0. Timing measured via `torch.cuda.Event` synchronization over 50k environment steps. Memory usage via `torch.cuda.max_memory_allocated()`.

The higher wall-clock overhead ($\approx 1.7\times$) relative to per-step overhead ($\approx 1.15\times$) is due to:

- Counterfactual forward passes for all actions during evaluation (not every training step).
- Periodic EMA target network updates (every 100 steps).
- Hebbian memory read operations during forecasting.

Memory overhead is modest (8–12%) due to:

- IAE state storage: $O(Nk) = O(16 \times 32) = 512$ floats.
- Hebbian matrices: $O(Nkd) = O(16 \times 32 \times 32) = 16K$ floats.
- EMA target network: Duplicate of predictor weights ($\approx 64K$ parameters).

For comparison, the policy and value networks together have $\approx 200K$ parameters, making ESAI-v3 additions a $< 50\%$ parameter increase.

I Appendix I: Extended Baseline Results

ESAI-v3 achieves consistent improvements across all environments. Gains are largest on coordination-heavy tasks (Social Distress: +13% vs CPO, Overcooked: +8% vs CPO) where graph diffusion and attention mechanisms enable robust multi-agent communication. Continuous-control MPE results demonstrate graceful extension beyond discrete domains, with smaller but statistically significant improvements (+5% vs CPO, $p = 0.003$).

J Appendix J: Statistical Test Summary

All primary comparisons use two-sample Welch's t-tests (unequal variance) with $\alpha = 0.05$ significance level. Key results:

All primary claims achieve $p < 0.01$ with large effect sizes ($d > 0.8$), indicating robust statistical support. We do not apply multiple-comparison correction (e.g., Bonferroni) because evaluations are across independent environments and hypotheses, not multiple tests of the same hypothesis.

Table 9: Complete hyperparameter settings for ESAI-v3 and all baselines.

Category	Parameter	Value
Optimization	Learning rate	3×10^{-4}
	Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
	Batch size	64
	Rollout horizon	2048
	PPO clip ϵ	0.2
	GAE λ	0.95
	Discount γ	0.99
	Entropy coefficient β	0.01 (annealed to 0.001)
	Max gradient norm	0.5
IAE Dynamics	Latent dimension k	32
	Persistence γ_E	0.9
	MLP hidden dims	[128, 128]
	Activation	ReLU
	Diffusion strength α	0.05
	Lipschitz constant L_g	1.0 (enforced via clipping)
Counterfactual	Alignment penalty λ_{reg}	0.1
	Neighbor weight κ	0.5
	Temperature τ_0	1.0
	Temperature $\tau_{\min}$	0.01
	Anneal steps K_τ	5×10^5
	EMA rate τ_{ema}	0.995
	Forecast network dims	[64, 64]
Hebbian	Learning rate η_H	1×10^{-3}
	Decay δ_H	0.02
	Regularizer λ_H	1×10^{-3}
	Memory dimension d	32
Diffusion	Neighborhood radius	2 (grid environments)
	Laplacian regularizer λ_D	1×10^{-2}
	Bias regularizer λ_{bias}	0.01 (swept: $\{0, 0.001, 0.01, 0.05, 0.1\}$)
	Spectral radius bound $\rho(L)$	≤ 0.95 (enforced)
	Identity embedding dim d_{id}	8
Training	Total environment steps	1×10^6
	Random seeds	5 (42, 123, 456, 789, 1011)
	GPU	NVIDIA A100 (40GB)
	Wall-clock time	≈ 6 hours per seed
	Evaluation frequency	Every 5×10^4 steps
	Evaluation episodes	100 per checkpoint

Table 10: Performance degradation under demonstration label noise.

Noise Level	PR	AR	APA
0% (clean)	0.82 ± 0.04	0.44 ± 0.03	0.76 ± 0.03
10%	0.79 ± 0.05	0.49 ± 0.04	0.72 ± 0.04
20%	0.73 ± 0.06	0.58 ± 0.05	0.65 ± 0.05
30%	0.51 ± 0.08	0.92 ± 0.09	0.48 ± 0.07
50%	0.38 ± 0.09	1.34 ± 0.12	0.34 ± 0.08

Table 11: ESAI-v3 behavior under culturally-variant demonstrations on Moral Temptation.

Demo Variant	Harm Threshold	PR	AR	Help Gap
Harsh (low tolerance)	0.1	0.89 ± 0.03	0.31 ± 0.04	0.62 ± 0.05
Neutral (balanced)	0.3	0.82 ± 0.04	0.44 ± 0.03	0.52 ± 0.04
Lenient (high tolerance)	0.5	0.71 ± 0.06	0.61 ± 0.05	0.41 ± 0.06

Table 12: Effect of temperature schedule on ESAI-v3 performance (Moral Temptation, $\Delta R = 5$).

Schedule	PR	AR	APA
Annealed ($1.0 \to 0.01$)	0.82 ± 0.04	0.44 ± 0.03	0.76 ± 0.03
Fixed low ($\tau = 0.01$)	0.71 ± 0.06	0.61 ± 0.05	0.68 ± 0.05
Fixed high ($\tau = 1.0$)	0.68 ± 0.07	0.58 ± 0.06	0.71 ± 0.04
No annealing (constant 0.5)	0.74 ± 0.06	0.54 ± 0.05	0.69 ± 0.05

Table 13: Prosocial ratio sensitivity to key hyperparameters. Bold indicates default value.

Parameter	Value 1	Value 2	Value 3	Value 4
γ_E	0.78 (0.7)	0.81 (0.85)	**0.82 (0.9)**	0.79 (0.95)
α	0.76 (0.01)	0.79 (0.03)	**0.82 (0.05)**	0.77 (0.1)
δ_H	0.74 (0.005)	0.79 (0.01)	**0.82 (0.02)**	0.80 (0.05)
k (IAE dim)	0.76 (16)	**0.82 (32)**	0.81 (64)	0.74 (128)
λ_{reg}	0.73 (0.01)	0.79 (0.05)	**0.82 (0.1)**	0.78 (0.5)

Table 14: Detailed computational profiling (mean over 3 independent runs).

Metric	Raw PPO	ESAI-v3	Overhead
Forward pass (ms/step)	2.3 ± 0.1	2.7 ± 0.1	+17.4%
Backward pass (ms/step)	3.1 ± 0.2	3.6 ± 0.2	+16.1%
Counterfactual forecast (ms)	—	1.2 ± 0.1	—
Diffusion update (ms)	—	0.3 ± 0.05	—
Wall-clock (50k steps, min)	18.2 ± 0.4	31.1 ± 0.6	+70.9%
Peak memory (GB)	6.8 ± 0.1	7.6 ± 0.2	+11.8%
FLOPs per step ($\times 10$)	8.1 ± 0.2	9.3 ± 0.3	+14.8%

Table 15: Full baseline comparison across all environments (prosocial ratio, mean $\pm$ std over 5 seeds).

Method	Moral Tempt	Social Dist	MPE	Overcooked
ESAI-v3	0.82 ± 0.04	0.84 ± 0.03	0.87 ± 0.03	0.82 ± 0.04
CPO	0.71 ± 0.04	0.76 ± 0.04	0.83 ± 0.04	0.76 ± 0.05
Reward Shaping	0.68 ± 0.05	0.71 ± 0.05	0.79 ± 0.05	0.71 ± 0.06
CIRL Proxy	0.65 ± 0.06	0.73 ± 0.05	0.81 ± 0.04	0.68 ± 0.07
Multi-Obj RL	0.69 ± 0.05	0.73 ± 0.04	0.83 ± 0.04	0.74 ± 0.05
Potential Shaping	0.66 ± 0.06	0.69 ± 0.06	0.77 ± 0.06	0.69 ± 0.07
Raw PPO	0.33 ± 0.09	0.39 ± 0.08	0.62 ± 0.07	0.51 ± 0.08

Table 16: Statistical comparison summary for primary metrics (ESAI-v3 vs baselines on Moral Temptation).

Comparison	Metric	t-statistic	p-value	Cohen's d
ESAI-v3 vs Raw PPO	PR	9.24	$< 10^{-6}$	2.41
ESAI-v3 vs CPO	PR	3.82	0.002	0.91
ESAI-v3 vs Reward Shaping	PR	4.21	0.001	1.03
ESAI-v3 vs Raw PPO	AR	-8.97	$< 10^{-6}$	-2.38
ESAI-v3 vs CPO	AR	-3.54	0.003	-0.84
Intervention (suppress vs baseline)	PR	7.31	$< 10^{-5}$	1.87
Intervention (amplify vs baseline)	PR	4.18	$< 10^{-3}$	1.12